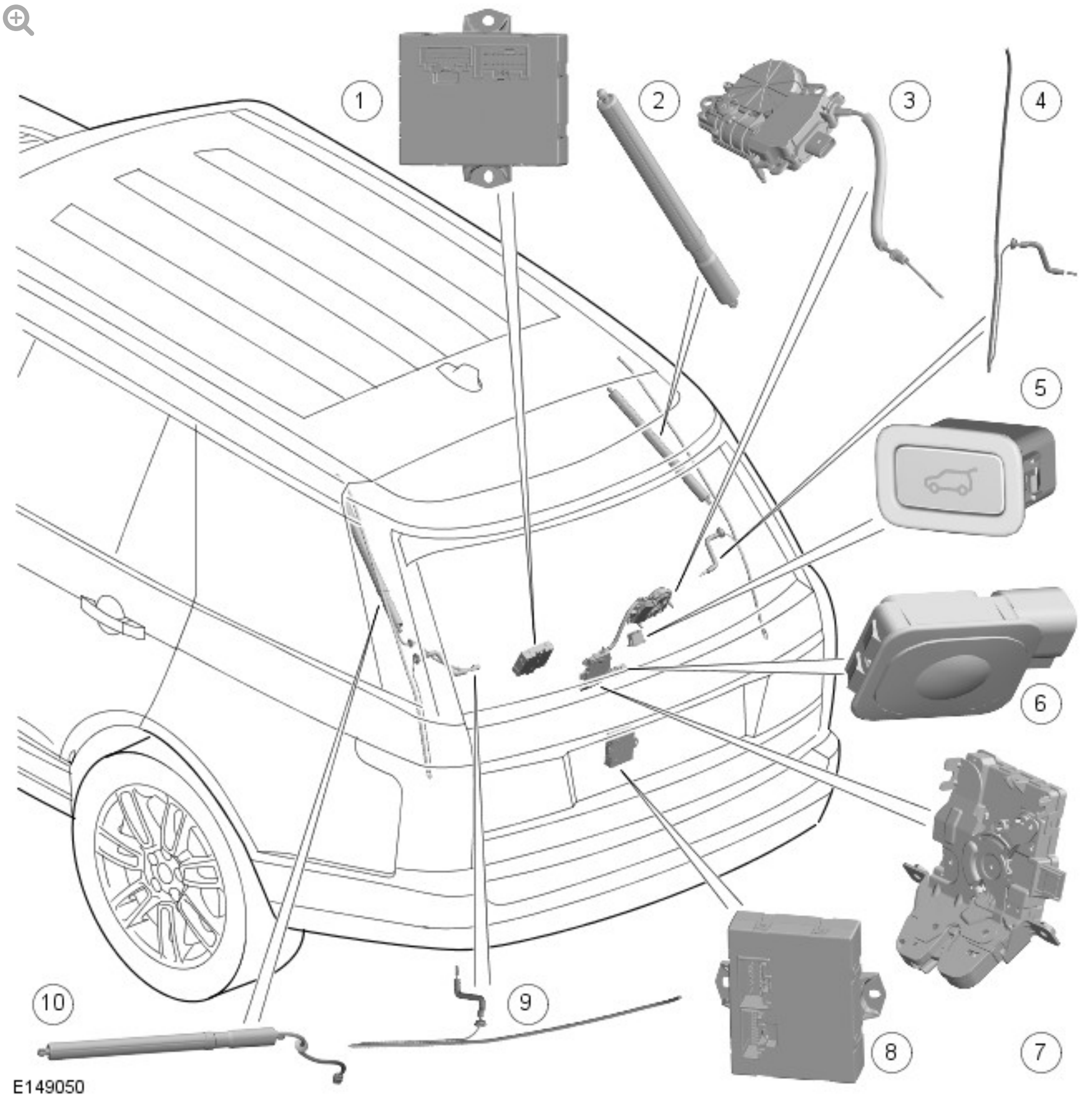
BODY CLOSURES

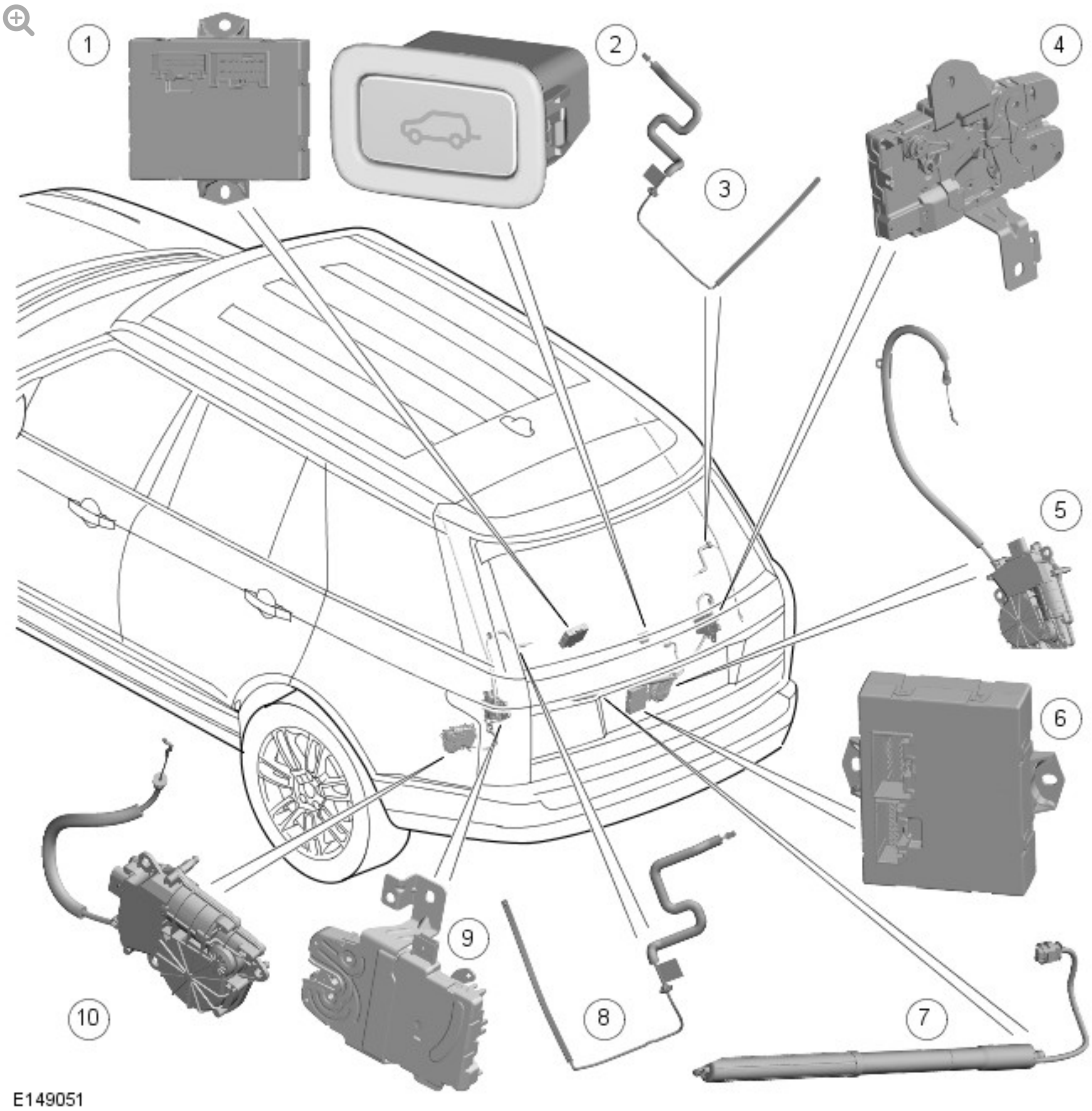
DESCRIPTION AND OPERATION

COMPONENT LOCATION – UPPER POWERED TAILGATE



ITEM	DESCRIPTION
1	Upper Tailgate Control Module (UTCM)
2	Right upper tailgate actuator
3	Upper tailgate soft close motor
4	Upper tailgate right anti-pinch strip
5	Tailgate global close switch
6	Upper tailgate open/close switch
7	Upper tailgate latch
8	Lower Tailgate Control Module (LTCM)
9	Upper tailgate left anti-pinch strip
10	Left upper tailgate actuator

COMPONENT LOCATION – LOWER POWERED TAILGATE



ITEM	DESCRIPTION
1	UTCM
2	Lower tailgate open/close switch
3	Lower tailgate right anti-pinch strip
4	Right lower tailgate latch
5	Right lower soft close actuator
6	LTCM
7	Lower tailgate actuator
8	Lower tailgate left anti-pinch strip
9	Left lower tailgate latch
10	Left lower soft close actuator

OVERVIEW

UPPER AND LOWER TAILGATES

The upper tailgate is constructed from lightweight plastic composites. The upper tailgate consists of three parts, which are bonded together to make the finished single piece tailgate. The outer body colored part is constructed from polypropylene, similar to a bumper construction to provide increased bump resistance.

The lower tailgate assembly is constructed from aluminum panels.

Depending on vehicle specification the tailgates are offered with either:

- manual opening and closing, or with a
- powered opening and closing function.

The upper powered tailgate is opened and closed by two electrically driven tailgate actuators mounted in place of the gas struts as used on the manual tailgate. The lower powered tailgate is opened and closed by one tailgate actuator also mounted in place of the gas strut.

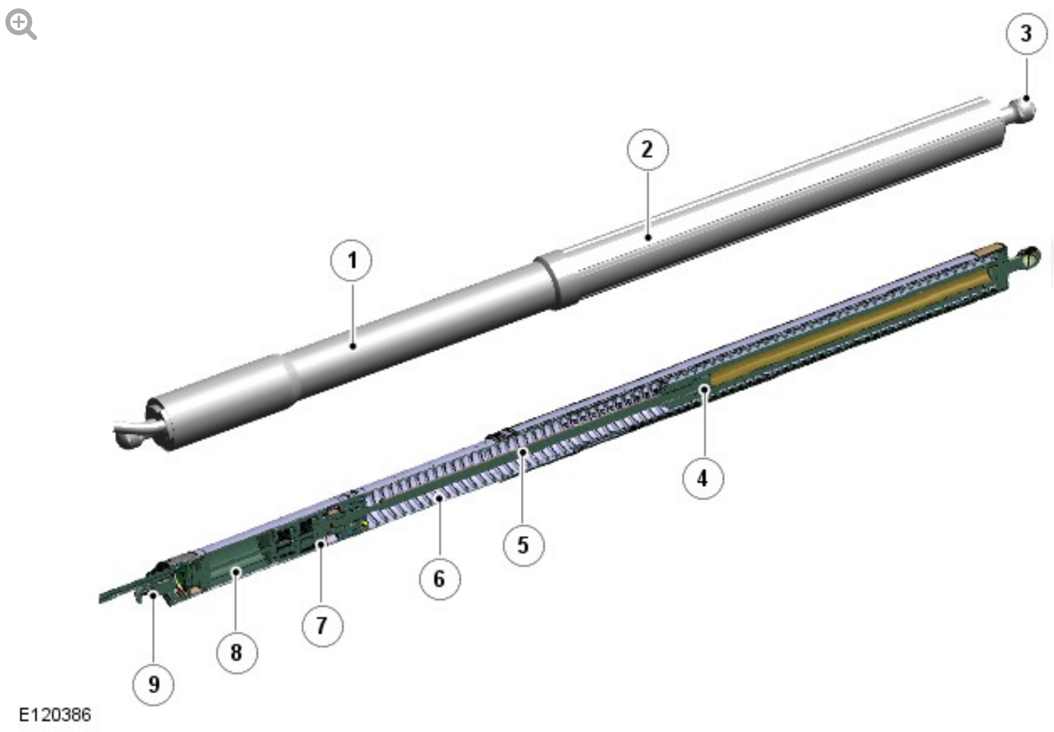
The tailgate actuator assembly has ball joints at either end allowing it to articulate between the fixed mounts on the vehicle. The nominal opening and closing time of the system is 5 seconds (+/2.0 seconds).

The tailgates operate using electronic latches with the addition of a soft-close closing motors to pull the latch closed for the last 6 mm of travel (soft close). The upper latches and soft closing motor are located in the upper tailgate. The lower latch and soft closing motors are located in the lower luggage compartment. The closing motors pull the latches closed at the end of the tailgates' travel, once the latches' secondary switch is reached.

DESCRIPTION

TAILGATE ACTUATORS

Tailgate actuator components (reference only)



ITEM	DESCRIPTION
1	Inner tube
2	Outer tube
3	Ball joint
4	Outer tube spindle nut
5	Spindle
6	Springs
7	Gear
8	Motor
9	Ball joint

The upper and lower tailgate actuators open and close the upper and lower tailgates using an electrically driven spindle located in the actuators' internal electric motor. Ball-joints positioned at each end of the actuator allow it to articulate between a fixed mount on the vehicle and the moveable tailgate. The tailgate opening operation is also aided by spring assistance.

The spindle drive comprises an inner and outer tube where the motor and gears in the inner tube drive a threaded spindle which runs on a threaded nut fixed to the inside of the outer tube.

The spindle drive incorporates an object detection function controlled by the tailgate modules. The function is similar to the anti-trap function of a closing electric window but operates in both directions. If the object detection feature is activated while the tailgate is closing, the spindle motor(s) stop and then reverse for a preset period. If the object detection feature is activated while a tailgate is opening, the spindle motor is stopped and held at that position.

NOTE:

> The spindle motor(s) on the tailgate experiencing an object/pinch condition whilst closing, will reverse to the fully open state.

A hall sensor, located in the spindle motor, monitors the speed of the motor. If the speed decreases below a set threshold, indicating an obstruction and increasing the motor current draw, the power feed to the motor is reversed causing the tailgate to move in the opposite direction of travel.

The amount of travel in the opposite direction, before stopping, is determined by the hall sensor count. An exception to the object detection reverse travel function occurs when the tailgate is opening through its first few degrees of travel from the latched position. In these circumstances, if an obstacle is detected, the tailgate stops in the position of the obstruction and no travel in the opposite direction occurs, other than a recoil action of the strut vs. hinge. The tailgate may recoil back onto the secondary latch but NO soft close shall occur.

POWERED TAILGATE

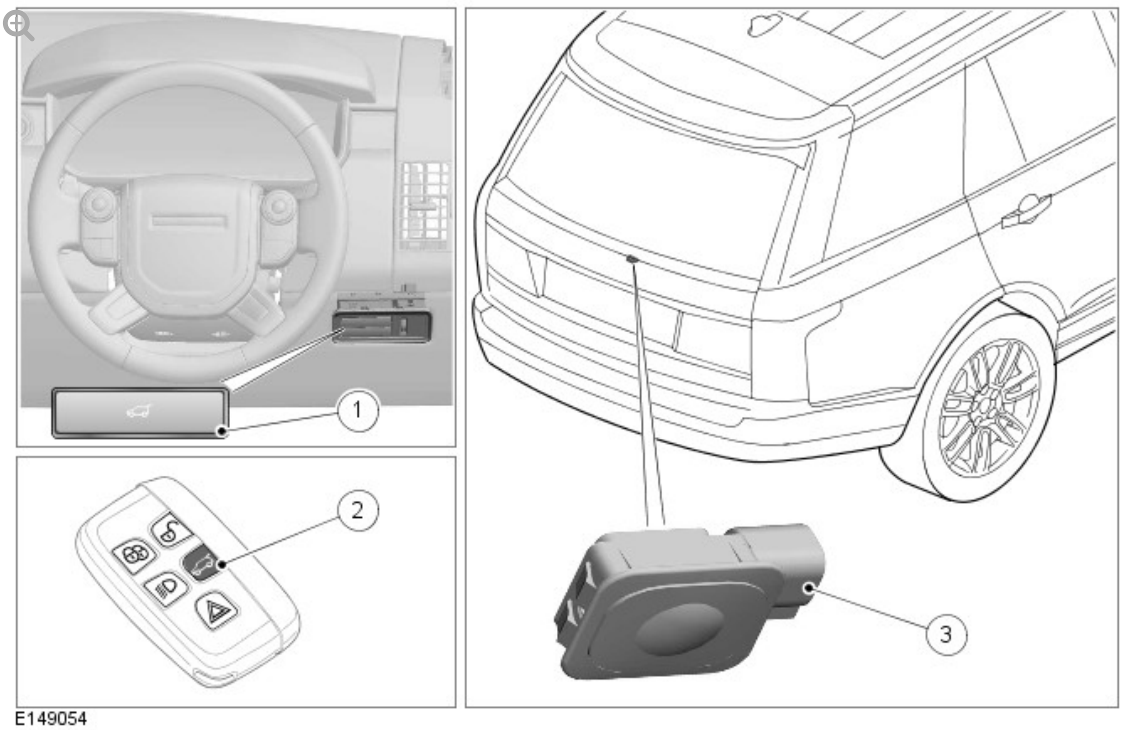
UPPER TAILGATE – RELEASE SWITCHES

The upper tailgate can be opened or closed using the following:

- The open/close switch on the smart key
- Rear license-plate-plinth, external tailgate open/close switch provided that the doors are unlocked and Park (P) is selected on the TCS.
- IC, internal tailgate open/close switch, providing:
  - the vehicle is not locked and alarmed
  - the vehicle speed is not at 5 kilometer/hour (3 mile/hour) or above.

These switches will also de-latch the manual upper tailgate:

- upper tailgate open/close switch
- upper tailgate open/close switch - Smart key
- upper tailgate open/close switch
- lower tailgate release switch.



ITEM	DESCRIPTION

1	Upper tailgate open/close switch
2	Upper tailgate open/close switch - Smart key
3	Upper tailgate open/close switch

The lower tailgate can be opened or closed using the following:

- lower tailgate open/close switch (located on the lower tailgate closing edge).

The powered tailgates can also be manually closed, to either the latches' secondary or primary position, without causing any damage to the power mechanism.

The upper tailgate (Global Close) close switch is located on the upper tailgate closing edge.

ITEM	DESCRIPTION
1	Tailgate global close switch

The lower tailgate open/close switch is located on the lower tailgate closing edge.

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ITEM	DESCRIPTION
1	Tailgate lower switch

The tailgate can be stopped at any time during the open or close cycle by a single press on any of the tailgate control switches. Thereafter:

- Pressing any ‘release’ control switch will authorize the upper tailgate to change the direction of movement (Toggle).
- Pressing the ‘close’ control switch will authorize the tailgate to continue or start closing.

For further information, see chart below.

The upper tailgate has a 'Garage Position' setting, where it is possible to set the maximum height to which the tailgate will open.

To set the required height:

- Open the tailgate to the position of the required height and ensure the tailgate is stationary for at least three seconds.
- Press and hold, the tailgate close switch.
- Successful height programming will be indicated by a long beep heard from the upper tailgate.
- Close the tailgate manually, or via the close switch, then open again to check that it opens to the programmed height.

The maximum opening height is now set. To reset the maximum opening height to full, repeat the process, but fully open the tailgate before pressing and holding the close switch.

The opening and closing control strategies of upper and lower tailgates are listed in the following chart:

STRATEGY	OPENING	CLOSING	STOP	START FROM STOPPED POSITION AFTER OPENING	START FROM STOPPED POSITION AFTER CLOSING	GARAGE POSITION - SET/RESET
Smart key (upper tailgate)	One push	One push	One push	One push for toggle closing	One push for toggle opening	X
Interior tailgate release (upper tailgate)	One push	One push	One push	One push for toggle closing	One push for toggle opening	X
Exterior tailgate release (upper	One push	One push	One push	One push for toggle closing	One push for toggle opening	X

tailgate)						
Global open/close switch (Upper and lower tailgates)	X	One push	One push	One push for closing	One push for closing	One push until long beep
Lower tailgate open/close switch	One push	One push	One push	One push for toggle closing	One push for toggle opening	X

If any object is detected that would interfere with the closing of either tailgate, the tailgate will stop and reverse to the fully open position. An audible mislock warning, that is 2 beeps from the security sounder, will indicate either a failed closing action or object or pinch detection. Any obstructions must be removed before pressing the close switch again.

The powered tailgate may lose its position memory if:

- there are multiple object detections
- there are inadvertent loads on the tailgate while it is moving; for example a person leaning against the tailgate
- the battery voltage is low
- any latch is manually closed.

The above may also inhibit the powered operation of the tailgate.

To reset the tailgate:

- Manually close the tailgate.
- Press a tailgate release switch.
- Allow the tailgate to power fully open or to the previously set position.
- Press and release the close switch.
- Allow the tailgate to power close fully.

The tailgate programmed position memory will now be restored.

NOTE:

If the vehicle is locked and the alarm is armed, if the tailgate is opened, and subsequently closed (via an authorized user/key) with the valid smart key, that was used to open the tailgate, left inside, the upper tailgate will re-open and an audible mislock warning, that is 2 beeps from the security sounder, will indicate a failed closing action. Please remove key and reclose tailgate. The smart key will not be detected if it is any way covered/obstructed by metal objects/laptops etc.

OPERATION

UPPER AND LOWER POWERED TAILGATES

CAUTION:

DO NOT CLOSE THE TAILGATE IF THE VEHICLE BATTERY IS DISCONNECTED. It is advisable to close the upper and lower latch claws when working on a vehicle with the battery disconnected to prevent accidental closure.

NOTE:

If required, the tailgate can be opened and closed manually.

The powered opening and closing actions are controlled by the tailgate modules. The modules receive permanent battery power supply from the RJB (rear junction box) and they are connected with each other via medium speed CAN and hardwired connection (anti-trap detection).

UPPER TAILGATE OPEN/CLOSE SEQUENCE

To initiate the upper tailgate opening sequence a 'tailgate release-request' is received by the CJB (central junction box) from one of the following:

- KVM - signal originates from the smart key tailgate open/close switch
- IC, internal tailgate open/close switch
- Rear license-plate-plinth, external tailgate open/close switch.

The CJB responds with the following simultaneous actions:

- A hardwire power release to the upper latch actuator mechanism
- A powered opening request-signal and latch status transmitted to the upper tailgate module via the medium speed CAN (controller area network) bus.

The processing of the opening signal by the CJB is influenced by a number of factors:

- Source of 'opening signal'
- Vehicle status: locked or unlocked
- Vehicle equipped with or without passive entry
- Vehicle speed and Transmission Control Switch (TCS) position.

Once the latch is released, the upper tailgate module actuates the motors located in tailgate actuator to raise the tailgate to its fully open or pre-set position. When the tailgate is in its opening cycle the automatic stop-position of the tailgate is functioned by a hall-sensor located in the motor. The hall-sensor signal transmitted to the tailgate module is synchronized with the pre-set memory indicating the stop position of the tailgate.

To close the tailgate (upper/lower) a hardwired signal is transmitted directly to the tailgate module (upper/lower) when the switch located on upper tailgates' closing edge is pressed and then released. The module operates the tailgate actuators in the opposite direction to close the tailgate to the secondary latched position. (The upper tailgate can be closed via the smart key/interior/exterior tailgate release buttons also.)

When the latch engages with the striker plate a signal transmitted from the latches' secondary switch, located in the latch mechanism, to the tailgate module confirms the latch is engaged. The module de-activates the tailgate actuators and activates the tailgate soft closing motor to pull the latch closed through the last 6mm of travel. This is the latches' primary fully-closed position. The tailgate soft closing motor then reverses to allow the latch to be opened upon the next release request. Mechanical Bowden cables connection between the motors and latch assemblies are used to complete the closing process of claw contraction and then reverse.

If the battery is disconnected when the tailgate is closed, the stored and calibrated tailgate opening heights will remain stored when the battery is reconnected. However, if power supply to the tailgate module is disconnected when the tailgate is open, the system will not recognize the tailgate's position, and subsequently will not function to any switch commands when power is reinstated. To reset the system the tailgate must be moved manually to the closed position where it will perform a soft-close and reset the hall-sensor counters to zero.

LOWER TAILGATE OPEN/CLOSE SEQUENCE

To initiate the lower tailgate opening sequence, the upper tailgate must first be fully opened. A 'lower tailgate release-request' is received by the CJB from the lower tailgate open/close switch.

The CJB responds with the following simultaneous actions:

- a hardwire power release to the lower latches actuator mechanisms
- a powered opening request-signal and latch status transmitted to the lower tailgate module via the medium speed CAN bus.

The processing of the lower opening signal by the CJB is influenced by a number of factors:

- source of 'opening signal'
- vehicle status: locked or unlocked
- vehicle equipped with or without passive entry
- vehicle speed and Transmission Control Switch (TCS) position.

Once the latches are released, the LTCM actuates the motors located in the lower tailgate actuator to lower the tailgate to its fully open position. When the lower tailgate is in its opening cycle the automatic stop-position of the tailgate is functioned by a hall-sensor located in the motor. The hall-sensor signal transmitted to the tailgate module is synchronized with the pre-set memory indicating the stop position of the tailgate.

To close the lower tailgate only, via the tailgate lower open/close switch, a hardwired signal is transmitted directly to the CJB when the switch located on lower tailgates' closing edge is pressed. The CJB responds with the following simultaneous actions:

- a hardwire power release to the lower latches actuator mechanisms
- a powered close request-signal and latch status transmitted to the lower tailgate module via the medium speed CAN (controller area network) bus.

To close the tailgate (upper/lower) via the global close switch a hardwired signal is transmitted directly to the UTCM/LTCM when the switch located on upper tailgates' closing edge is pressed and then released.

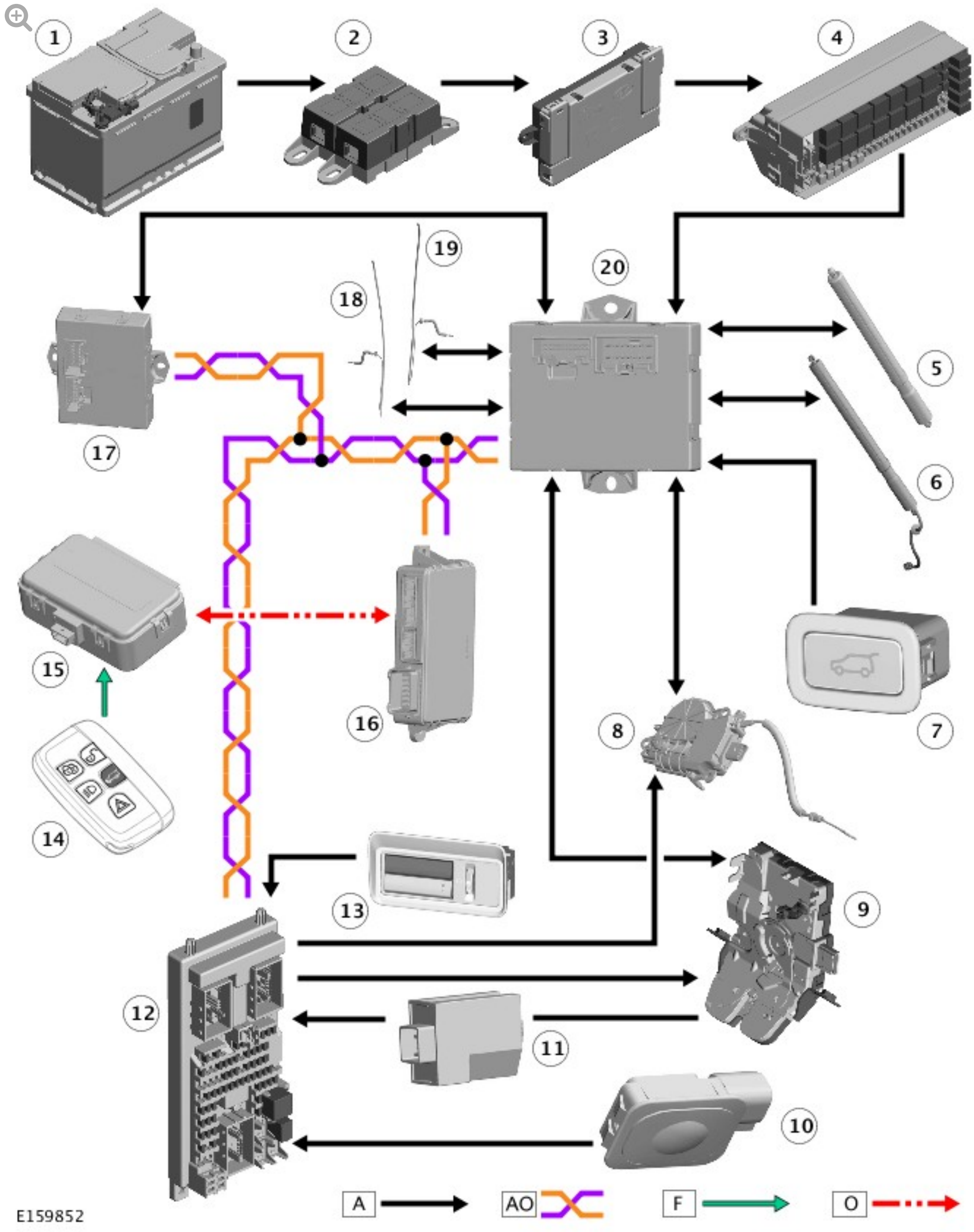
The LTCM operates the lower tailgate actuator in the opposite direction to close the lower tailgate to the secondary latched position.

When the latches engage with the striker(s) plate(s) a signal transmitted from the latches’ secondary switch, located in the latch mechanism, to the LTCM confirms the latches are engaged. The LTCM de-activates the lower tailgate actuator and activates the lower tailgate soft closing motors to pull the latches closed through the last 6mm of travel. This is the latches' primary fully-closed position. The tailgate soft closing motors then reverse to allow the latches to be opened upon the next release request.

Mechanical Bowden cables connection between the motors and latch assemblies are used to complete the closing process of claw contraction and then reverse.

If the battery is disconnected when the tailgate is closed, the calibrated tailgate opening height will remain stored when the battery is reconnected. However, if power supply to the tailgate module is disconnected when the tailgate is open, the system will not recognize the tailgate's position, and subsequently will not function to any switch commands when power is reinstated. To reset the system the tailgate must be moved manually to the closed position where it will perform a soft-close and reset the hall-sensor counters to zero.

CONTROL DIAGRAM – UPPER POWERED TAILGATE

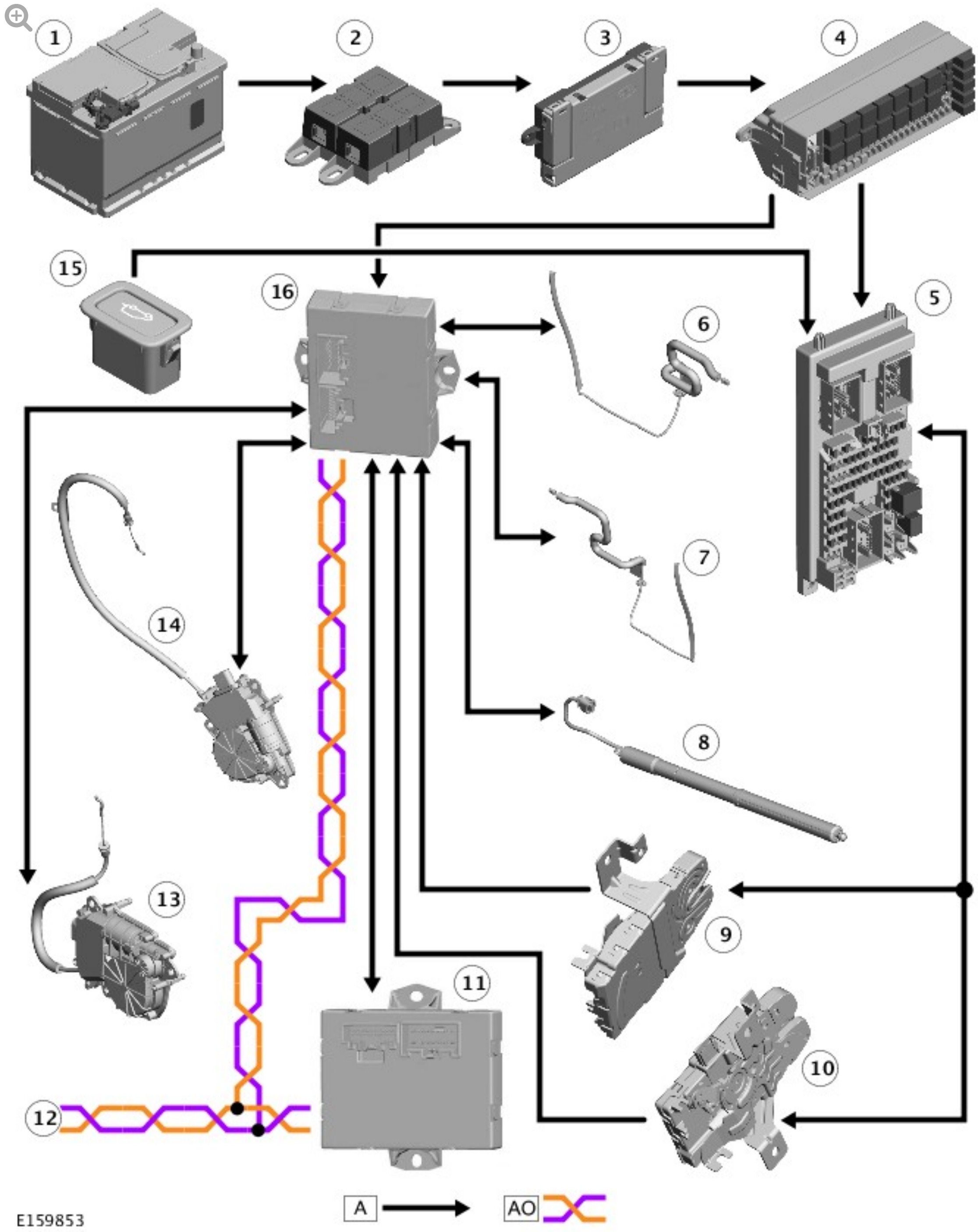


A = HARDWIRED; AO = MEDIUM SPEED CAN BODY SYSTEMS; F = RF (RADIO FREQUENCY) TRANSMISSION.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB) 2
3	BJB
4	Rear Junction Box (RJB)

5	Right upper tailgate actuator
6	Left upper tailgate actuator
7	Tailgate global close switch
8	Upper soft close actuator
9	Upper tailgate latch
10	Tailgate exterior open/close switch
11	RF filter
12	Central Junction Box (CJB)
13	Tailgate interior open/close switch
14	Tailgate open/close switch - smart key
15	RF receiver
16	Keyless Vehicle Module (KVM)
17	LTCM
18	Upper tailgate left anti-pinch strip
19	Upper tailgate right anti-pinch strip
20	UTCM

CONTROL DIAGRAM – LOWER POWERED TAILGATE



A = HARDWIRED; AO = MEDIUM SPEED CAN BODY SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	BJB 2
3	BJB
4	RJB

5	CJB
6	Lower tailgate right anti-pinch strip
7	Lower tailgate right anti-pinch strip
8	Lower tailgate actuator
9	Right lower tailgate latch
10	Left lower tailgate latch
11	UTCM
12	CAN connection to other systems
13	Left lower soft close actuator
14	Right lower soft close actuator
15	Tailgate lower open/close switch
16	LTCM